

Income and Wealth Distribution in Macroeconomics

A Continuous-Time Approach

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Motivation

- Key development over last 30 years: incorporation of explicit **heterogeneity** into macro models
- Welcome development because:
 1. can bring **micro data** to table to discipline **macro theories**
 2. can talk about **welfare** implications of shocks, policies
 3. **aggregate implications** often **differ** from rep agent models
- Despite increasing popularity of heterogeneous agent models:
 1. very **few theoretical results**, almost everything numerical
 2. even numerical analyses can be **difficult, costly**

This Paper: solving het. agent model = solving PDEs

- We recast Aiyagari-Bewley-Huggett model in **continuous time**
⇒ boils down to **system of PDEs**
- Take advantage of this to make two types of contributions:
- **New theoretical results:**
 1. analytics: consumption, saving, MPCs of the poor
 2. closed-form for wealth distribution with 2 income types
 3. unique stationary equilibrium if $IES \geq 1$ (sufficient condition)
 4. characterization of “soft” borrowing constraints
- **Computational algorithm:**
 - simple, efficient (think 0.25 seconds), portable
 - particularly well-suited for problems with **non-convexities** ...
 - ... and **transition dynamics**
 - codes: <http://www.princeton.edu/~moll/HACTproject.htm>

Solving het. agent model = solving PDEs

- More precisely: a system of two PDEs
 1. **Hamilton-Jacobi-Bellman** equation for individual choices
 2. **Kolmogorov Forward** equation for evolution of distribution
- Many well-developed methods for analyzing and solving these
- Apparatus is very **general**: applies to **any** heterogeneous agent model with continuum of atomistic agents
 1. heterogeneous households (Aiyagari, Bewley, Huggett,...)
 2. heterogeneous producers (Hopenhayn,...)
- can be extended to handle aggregate shocks (Krusell-Smith,...)
 - “When Inequality Matters for Macro and Macro Matters for Inequality” (with Ahn, Kaplan, Winberry & Wolf)

Workhorse Model of Income and Wealth Distribution in Macroeconomics

Workhorse Model of Income and Wealth Distribution

Households are heterogeneous in their wealth a and income y , solve

$$\max_{\{c_t\}_{t \geq 0}} \mathbb{E}_0 \int_0^{\infty} e^{-\rho t} u(c_t) dt \quad \text{s.t.}$$

$$\dot{a}_t = y_t + r a_t - c_t$$

$$y_t \in \{y_1, y_2\} \text{ Poisson with intensities } \lambda_1, \lambda_2$$

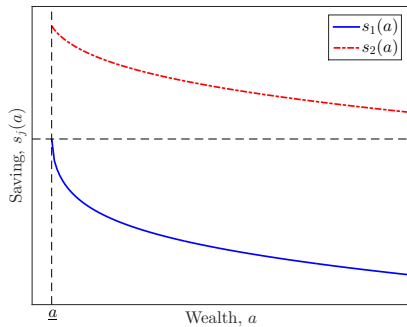
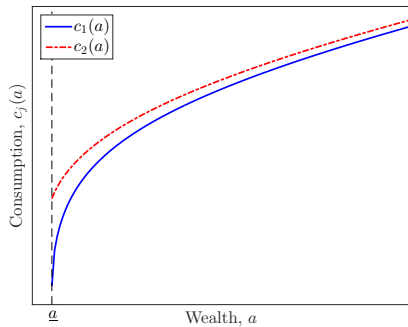
$$a_t \geq \underline{a}$$

- c_t : consumption
- u : utility function, $u' > 0$, $u'' < 0$
- ρ : discount rate
- r_t : interest rate
- $\underline{a} \geq -y_1/r$: borrowing limit e.g. if $\underline{a} = 0$, can only save

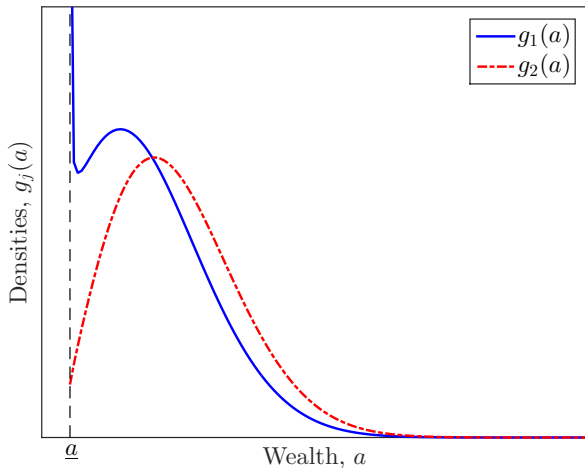
Later: carries over to y_t = more general processes, e.g. diffusion

Equilibrium (Huggett): bonds in fixed supply, i.e. aggregate a_t = fixed

Typical Consumption and Saving Policy Functions



Typical Stationary Distribution



Equations for Stationary Equilibrium

$$\rho v_j(a) = \max_c u(c) + v'_j(a)(y_j + ra - c) + \lambda_j(v_{-j}(a) - v_j(a)) \quad (\text{HJB})$$

$$0 = -\frac{d}{da}[s_j(a)g_j(a)] - \lambda_j g_j(a) + \lambda_{-j} g_{-j}(a), \quad (\text{KF})$$

$s_j(a) = y_j + ra - c_j(a)$ = saving policy function from (HJB),

$$\int_{\underline{a}}^{\infty} (g_1(a) + g_2(a)) da = 1, \quad g_1, g_2 \geq 0$$

$$S(r) := \int_{\underline{a}}^{\infty} a g_1(a) da + \int_{\underline{a}}^{\infty} a g_2(a) da = B, \quad B \geq 0 \quad (\text{EQ})$$

- The two PDEs (HJB) and (KF) together with (EQ) fully characterize stationary equilibrium

► Derivation of (HJB)

► (KF)

Transition Dynamics

- Needed whenever initial condition $\neq$ stationary distribution
- Equilibrium still coupled systems of HJB and KF equations...
- ... but now **time-dependent**: $v_j(a, t)$ and $g_j(a, t)$
- See paper for equations
- Difficulty: the two PDEs run in opposite directions in time
 - HJB looks forward, runs backwards from terminal condition
 - KF looks backward, runs forward from initial condition

Borrowing Constraints?

- Q: where is borrowing constraint $a \geq \underline{a}$ in (HJB)?
- A: “in” boundary condition
- **Result:** v_j must satisfy

$$v_j'(\underline{a}) \geq u'(y_j + r\underline{a}), \quad j = 1, 2 \quad (\text{BC})$$

- **Derivation:**
 - the FOC still holds at the borrowing constraint

$$u'(c_j(\underline{a})) = v_j'(\underline{a}) \quad (\text{FOC})$$

- for borrowing constraint not to be violated, need

$$s_j(\underline{a}) = y_j + r\underline{a} - c_j(\underline{a}) \geq 0 \quad (*)$$

- (FOC) and (*) $\Rightarrow$ (BC).
- See slides on viscosity solutions for more rigorous discussion

Plan

- New theoretical results:

1. analytics: consumption, saving, MPCs of the poor
2. closed-form for wealth distribution with 2 income types
3. unique stationary equilibrium if $IES \geq 1$ (sufficient condition)
4. “soft” borrowing constraints

Note: for 1., 2. and 4. analyze **partial equilibrium** with $r < \rho$

- Computational algorithm:

- problems with non-convexities
- transition dynamics

Result 1: Consumption, Saving Behavior of the Poor

Consumption/saving behavior near borrowing constraint depends on:

1. tightness of constraint
2. properties of u as $c \rightarrow 0$

Assumption 1:

As $a \rightarrow \underline{a}$, coefficient of absolute risk aversion $R(c) := -u''(c)/u'(c)$ remains finite

$$-\frac{u''(y_1 + r\underline{a})}{u'(y_1 + r\underline{a})} < \infty$$

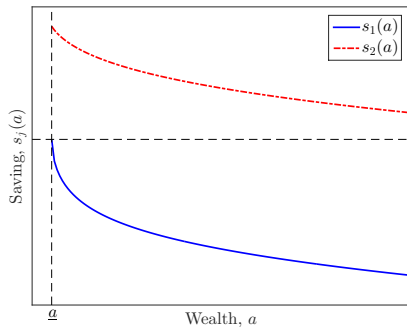
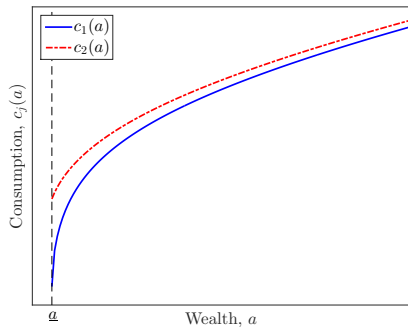
- will show: A1 $\Rightarrow$ borrowing constraint “matters” (in fact, it’s an $\Leftrightarrow$)

How to read A1?

- “standard” utility functions, e.g. CRRA, satisfy $-\frac{u''(0)}{u'(0)} = \infty$
- hence for standard utility functions A1 equivalent to $\underline{a} > -y_1/r$, i.e. constraint matters if it is tighter than “natural borrowing constraint”
- but weaker: e.g. if $u'(c) = e^{-\theta c}$, constraint matters even if $\underline{a} = -\frac{y_1}{r}$

Result 1: Consumption, Saving Behavior of the Poor

Rough version of Proposition: under A1 policy functions look like this



Result 1: Consumption, Saving Behavior of the Poor

Proposition: Assume $r < \rho$, $y_1 < y_2$ and that A1 holds.

Then saving and consumption policy functions close to $a = \underline{a}$ satisfy

$$s_1(a) \sim -\sqrt{2\nu_1} \sqrt{a - \underline{a}}$$

$$c_1(a) \sim y_1 + ra + \sqrt{2\nu_1} \sqrt{a - \underline{a}}$$

$$c'_1(a) \sim r + \frac{1}{2} \sqrt{\frac{\nu_1}{2(a - \underline{a})}}$$

where $\nu_1 = \text{constant}$ that depends on $r, \rho, \lambda_1, \lambda_2$ etc – see next slide

Note: “ $f(a) \sim g(a)$ ” means $\lim_{a \rightarrow \underline{a}} f(a)/g(a) = 1$, “ f behaves like g close to $\underline{a}$ ”

Result 1: Consumption, Saving Behavior of the Poor

Corollary: The wealth of worker who keeps y_1 converges to borrowing constraint in finite time at speed governed by ν_1 :

$$a(t) - \underline{a} \sim \frac{\nu_1}{2} (T - t)^2, \quad T := \text{"hitting time"} = \sqrt{\frac{2(a_0 - \underline{a})}{\nu_1}}, \quad 0 \leq t \leq T$$

Proof: integrate $\dot{a}(t) = -\sqrt{2\nu_1} \sqrt{a(t) - \underline{a}}$

And have analytic solution for speed

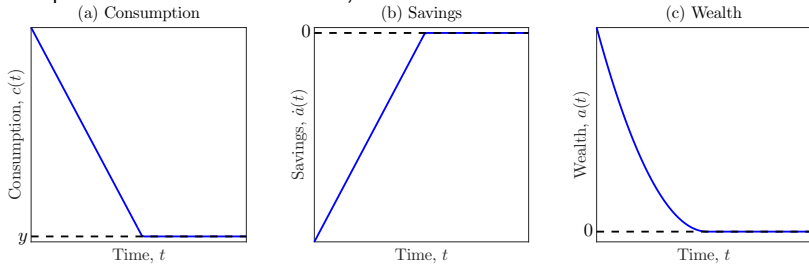
$$\begin{aligned} \nu_1 &= \frac{(\rho - r)u'(\underline{c}_1) + \lambda_1(u'(\underline{c}_1) - u'(\underline{c}_2))}{-u''(\underline{c}_1)} \\ &\approx (\rho - r)\text{IES}(\underline{c}_1)\underline{c}_1 + \lambda_1(\underline{c}_2 - \underline{c}_1) \end{aligned}$$

Intuition for Result 1: Two Special Cases

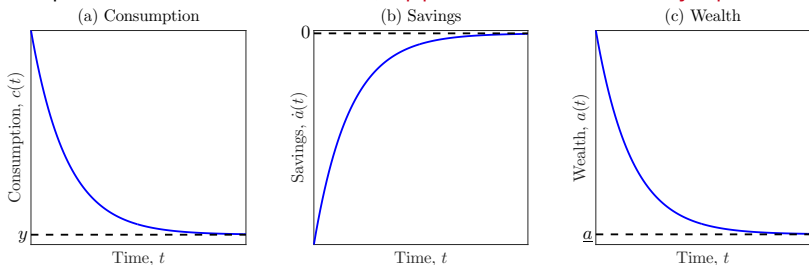
- What's the role of A_1 ? And why the square root?
- Explain using two special cases with **analytic solution**
- Both cases: **no income uncertainty**

Intuition for Result 1: Two Special Cases

- Special case 1: A1 holds, **hit** constraint



- Special case 2: A1 violated, **approach** constraint **asymptotically**



Intuition for Result 1: Two Special Cases

Special case 1: **hit** constraint

- exponential utility $u'(c) = e^{-\theta c}$, tight constraint

$$\dot{c} = \frac{1}{\theta}(r - \rho), \quad \dot{a} = y + ra - c, \quad a \geq 0$$

- satisfies A1: $-\frac{u''(y)}{u'(y)} = \theta < \infty$. Solution:

$$c(t) = y + \nu(\textcolor{blue}{T} - t), \quad a(t) = \frac{\nu}{2}(\textcolor{blue}{T} - t)^2, \quad \nu := \frac{\rho - r}{\theta}$$

Special case 2: only **approach** constraint **asymptotically**

- CRRA utility $u'(c) = c^{-\gamma}$, loose constraint

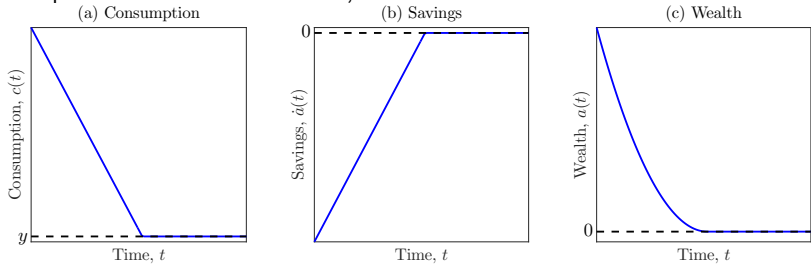
$$\frac{\dot{c}}{c} = \frac{1}{\gamma}(r - \rho), \quad \dot{a} = y + ra - c, \quad a \geq \underline{a} = -\frac{y}{r}$$

- violates A1: $-\frac{u''(y+ra)}{u'(y+ra)} \rightarrow \infty$ as $a \rightarrow \underline{a}$. Solution:

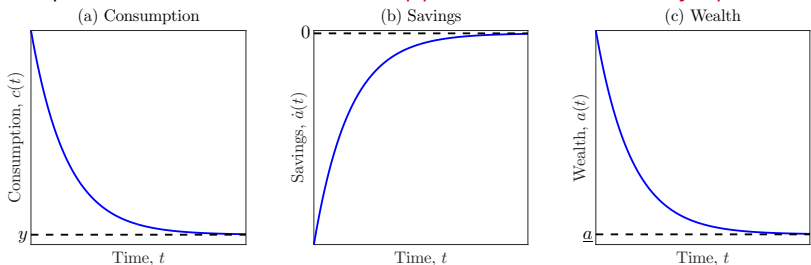
$$c(t) = y + (r + \eta)a(t), \quad a(t) - \underline{a} = (a_0 - \underline{a})e^{-\eta t}, \quad \eta := \frac{\rho - r}{\gamma}$$

Intuition for Result 1: Two Special Cases

- Special case 1: A1 holds, **hit** constraint



- Special case 2: A1 violated, **approach** constraint **asymptotically**



Consumption, Saving Behavior of the Rich

- Skip this today. See paper.

Marginal Propensities to Consume and Save

- So far: have characterized $c'_j(a) \neq \text{MPC over discrete time interval}$
- **Definition:** The MPC over a time period τ is given by

$$\text{MPC}_{j,\tau}(a) = C'_{j,\tau}(a), \quad \text{where}$$

$$C_{j,\tau}(a) = \mathbb{E} \left[\int_0^\tau c_j(a_t) dt \mid a_0 = a, y_0 = y_j \right]$$

- **Lemma:** If τ sufficiently small so that no income switches, then

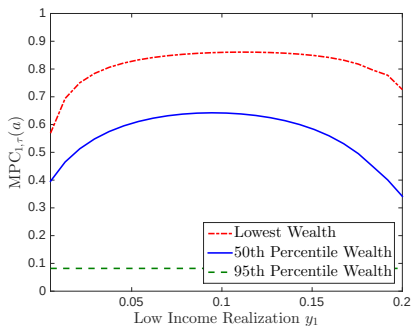
$$\text{MPC}_{1,\tau}(a) \sim \min\{\tau c'_1(a), 1 + \tau r\}$$

Note: $\text{MPC}_{1,\tau}(a)$ bounded above even though $c'_1(a) \rightarrow \infty$ as $a \downarrow \underline{a}$

- If new income draws before τ , no more analytic solution
- But straightforward computation using **Feynman-Kac formula**

Using the Formula for ν_1 to Better Understand MPCs

- Consider dependence of low-income type's $MPC_{1,\tau}(a)$ on y_1



- Why hump-shaped?!? Answer: $MPC_{1,\tau}(a)$ proportional to

$$c'_1(a) \sim r + \frac{1}{2} \sqrt{\frac{\nu_1}{2(a - \underline{a})}}, \quad \nu_1 \approx (\rho - r) \frac{1}{\gamma} \underline{c}_1 + \lambda_1 (\underline{c}_2 - \underline{c}_1)$$

and note that $\underline{c}_1 = y_1 + r\underline{a}$

- Can see: increase in y_1 has two offsetting effects

Result 2: Stationary Wealth Distribution

- Recall equation for stationary distribution

$$0 = -\frac{d}{da}[s_j(a)g_j(a)] - \lambda_j g_j(a) + \lambda_{-j} g_{-j}(a) \quad (\text{KF})$$

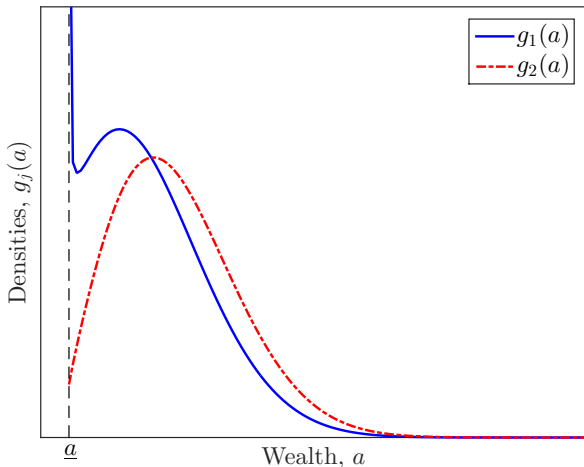
- Lemma:** the solution to (KF) is

$$g_i(a) = \frac{\kappa_j}{s_j(a)} \exp \left(- \int_{\underline{a}}^a \left(\frac{\lambda_1}{s_1(x)} + \frac{\lambda_2}{s_2(x)} dx \right) \right)$$

with κ_1, κ_2 pinned down by g_j 's integrating to one

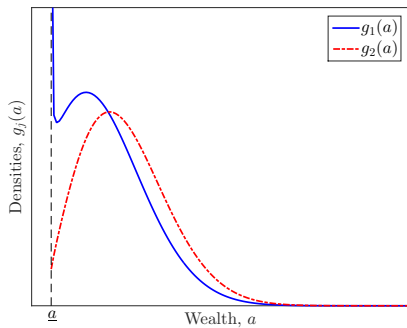
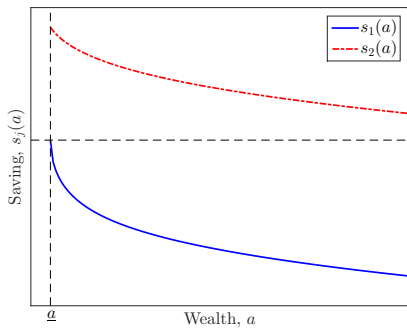
- Features of wealth distribution:
 - Dirac **point mass** of type y_1 individuals at constraint $G_1(\underline{a}) > 0$
 - thin right tail:** $g(a) \sim \xi(a_{\max} - a)^{\lambda_2/\zeta_2 - 1}$, i.e. not Pareto
 - see paper for more
- Later in paper: extension with Pareto tail (Benhabib-Bisin-Zhu)

Result 2: Stationary Wealth Distribution

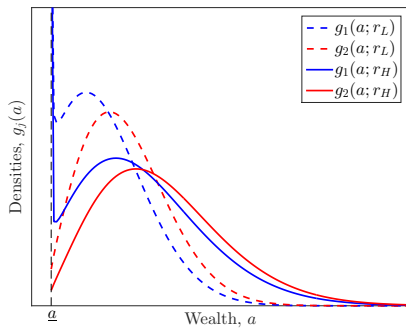
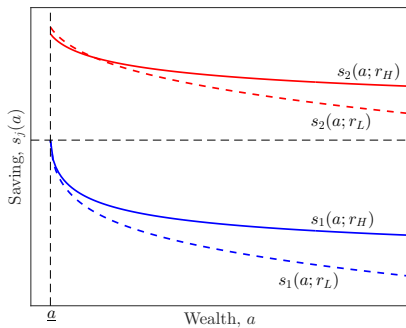


Note: in numerical solution, Dirac mass = finite spike in density

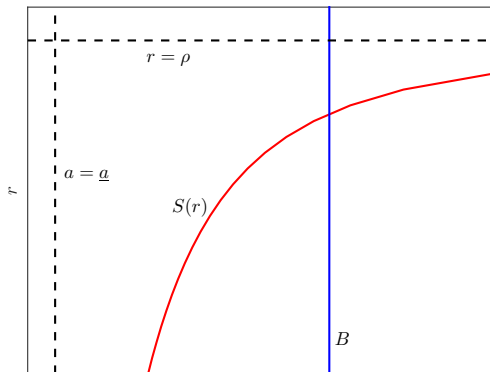
General Equilibrium: Existence and Uniqueness



Increase in r from r_L to $r_H > r_L$



Stationary Equilibrium



$$\text{Asset Supply } S(r) = \int_{\underline{a}}^{\infty} ag_1(a; r)da + \int_{\underline{a}}^{\infty} ag_2(a; r)da$$

- **Proposition:** a stationary equilibrium exists

Result 3: Uniqueness of Stationary Equilibrium

Proposition: Assume that the IES is weakly greater than one

$$\text{IES}(c) := -\frac{u'(c)}{u''(c)c} \geq 1 \quad \text{for all } c \geq 0,$$

and that there is no borrowing $a \geq 0$. Then:

1. Individual consumption $c_j(a; r)$ is strictly **d**ecreasing in r
2. Individual saving $s_j(a; r)$ is strictly **i**ncreasing in r
3. $r \uparrow \Rightarrow$ CDF $G_j(a; r)$ **s**hifts **r**ight in FOSD sense
4. Aggregate saving $S(r)$ is strictly **i**ncreasing $\Rightarrow$ **u**niqueness

Note: holds for **a**ny labor income process, not just two-state Poisson

Uniqueness: Proof Sketch

- Parts 2 to 4 direct consequences of part 1 ($c_j(a; r)$ decreasing in r)
- $\Rightarrow$ focus on part 1: builds on nice result by Olivi (2017) who decomposes $\partial c_j / \partial r$ into income and substitution effects
- **Lemma** (Olivi, 2017): c response to change in r is

$$\frac{\partial c_j(a)}{\partial r} = \underbrace{\frac{1}{u''(c_0)} \mathbb{E}_0 \int_0^T e^{-\int_0^t \xi_s ds} u'(c_t) dt}_{\text{substitution effect} < 0} + \underbrace{\frac{1}{u''(c_0)} \mathbb{E}_0 \int_0^T e^{-\int_0^t \xi_s ds} u''(c_t) a_t \partial_a c_t dt}_{\text{income effect} > 0}$$

where $\xi_t := \rho - r + \partial_a c_t$ and $T := \inf\{t \geq 0 | a_t = 0\}$ = time at which hit 0

- We show: $\text{IES}(c) := -\frac{u'(c)}{u''(c)c} \geq 1 \Rightarrow$ substitution effect dominates $\Rightarrow \partial c_j(a) / \partial r < 0$, i.e. consumption decreasing in r

Result 4: “Soft” Borrowing Constraints

- Empirical wealth distributions:
 1. individuals with **positive** wealth
 2. individuals with **negative** wealth
 3. **spike at** close to **zero** net worth
- Does not square well with Aiyagari-Bewley-Huggett model
- Simple solution: “**soft**” borrowing constraint = **wedge** between borrowing and saving r
- Paper: **first theoretical characterization** of “soft” constraint
 - square root formulas
 - Dirac mass at zero net worth

Computations for Heterogeneous Agent Model

Computational Advantages relative to Discrete Time

1. **Borrowing constraints** only show up in **boundary conditions**
 - FOCs always hold with “=”
2. **“Tomorrow is today”**
 - FOCs are “static”, compute by hand: $c^{-\gamma} = v'_j(a)$
3. **Sparsity**
 - solving Bellman, distribution = inverting matrix
 - but matrices very sparse (“tridiagonal”)
 - reason: continuous time $\Rightarrow$ one step left or one step right
4. **Two birds with one stone**
 - tight link between solving (HJB) and (KF) for distribution
 - matrix in discrete (KF) is **transpose** of matrix in discrete (HJB)
 - reason: diff. operator in (KF) is **adjoint** of operator in (HJB)

Computations for Heterogeneous Agent Model

- **Hard part:** HJB equation
- **Easy part:** KF equation. Once you solved HJB equation, get KF equation “for free”
- System to be solved

$$\rho v_1(a) = \max_c u(c) + v_1'(a)(y_1 + ra - c) + \lambda_1(v_2(a) - v_1(a))$$

$$\rho v_2(a) = \max_c u(c) + v_2'(a)(y_2 + ra - c) + \lambda_2(v_1(a) - v_2(a))$$

$$0 = -\frac{d}{da}[s_1(a)g_1(a)] - \lambda_1 g_1(a) + \lambda_2 g_2(a)$$

$$0 = -\frac{d}{da}[s_2(a)g_2(a)] - \lambda_2 g_2(a) + \lambda_1 g_1(a)$$

$$1 = \int_{\underline{a}}^{\infty} g_1(a) da + \int_{\underline{a}}^{\infty} g_2(a) da$$

$$B = \int_{\underline{a}}^{\infty} a g_1(a) da + \int_{\underline{a}}^{\infty} a g_2(a) da := S(r)$$

Bird's Eye View of Algorithm for Stationary Equilibria

- Use **finite difference method**:

<http://www.princeton.edu/~moll/HACTproject.htm>

- Discretize state space $a_i, i = 1, \dots, I$ with step size Δa

$$v'_j(a_i) \approx \frac{v_{i+1,j} - v_{i,j}}{\Delta a} \quad \text{or} \quad \frac{v_{i,j} - v_{i-1,j}}{\Delta a}$$

Denote $\mathbf{v} = \begin{bmatrix} v_1(a_1) \\ \vdots \\ v_2(a_I) \end{bmatrix}$, $\mathbf{g} = \begin{bmatrix} g_1(a_1) \\ \vdots \\ g_2(a_I) \end{bmatrix}$, dimension = $2I \times 1$

- End product of FD method: system of **sparse matrix equations**

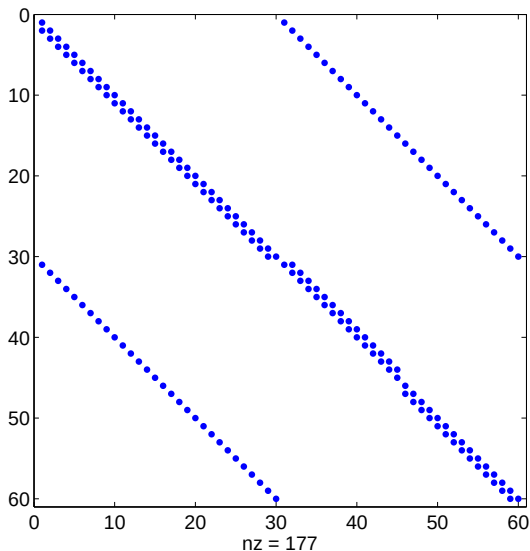
$$\rho \mathbf{v} = \mathbf{u}(\mathbf{v}) + \mathbf{A}(\mathbf{v}; r) \mathbf{v}$$

$$\mathbf{0} = \mathbf{A}(\mathbf{v}; r)^T \mathbf{g}$$

$$B = S(\mathbf{g}; r)$$

which is easy to solve on computer

Visualization of $\mathbf{A}$ (output of `spy(A)` in Matlab)



HJB Equation: Barles-Souganidis

- There is a well-developed theory for numerical solution of HJB equation using finite difference methods
- Key paper: Barles and Souganidis (1991), “Convergence of approximation schemes for fully nonlinear second order equations
- **Result:** finite difference scheme “converges” to unique viscosity solution under three conditions
 1. monotonicity
 2. consistency
 3. stability
- Good reference: Tourin (2013), “An Introduction to Finite Difference Methods for PDEs in Finance”
- Background on viscosity soln’s: “Viscosity Solutions for Dummies”
http://www.princeton.edu/~moll/viscosity_slides.pdf
- Accuracy? ▶ Two experiments, more in next revision – suggestions?

Transition Dynamics

- Natural generalization of algorithm for stationary equilibrium
 - denote $v_{i,j}^n = v_i(a_j, t^n)$ and stack into $\mathbf{v}^n$
 - denote $g_{i,j}^n = g_i(a_j, t^n)$ and stack into $\mathbf{g}^n$
- System of **sparse matrix equations** for transition dynamics:

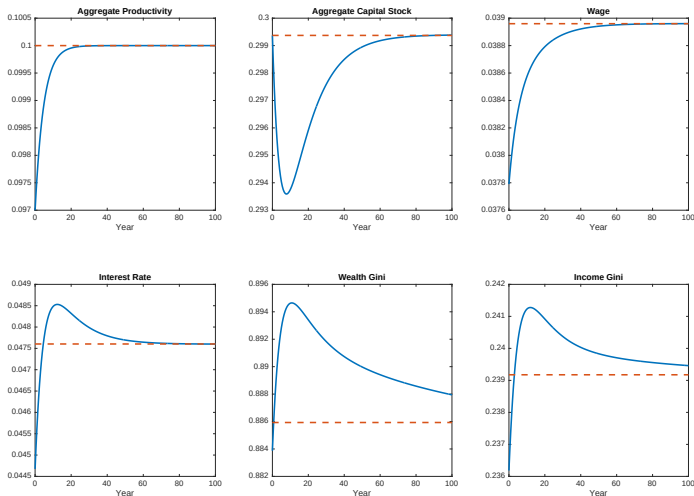
$$\begin{aligned}\rho \mathbf{v}^n &= \mathbf{u}(\mathbf{v}^{n+1}) + \mathbf{A}(\mathbf{v}^{n+1}; r^n) \mathbf{v}^n + \frac{\mathbf{v}^{n+1} - \mathbf{v}^n}{\Delta t}, \\ \frac{\mathbf{g}^{n+1} - \mathbf{g}^n}{\Delta t} &= \mathbf{A}(\mathbf{v}^n; r^n)^\top \mathbf{g}^{n+1}, \\ B &= S(\mathbf{g}^n; r^n),\end{aligned}$$

- Terminal condition for $\mathbf{v}$: $\mathbf{v}^N = \mathbf{v}_\infty$ (steady state)
- Initial condition for $\mathbf{g}$: $\mathbf{g}^1 = \mathbf{g}_0$.

An MIT Shock in the Aiyagari Model

- Production: $Y_t = F_t(K, L) = A_t K^\alpha L^{1-\alpha}$, $dA_t = \nu(\bar{A} - A_t)dt$

http://www.princeton.edu/~moll/HACTproject/aiyagari_poisson_MITshock.m



Generalizations and Other Applications

A Model with a Continuum of Income Types

- Assume idiosyncratic income follows diffusion process

$$dy_t = \mu(y_t)dt + \sigma(y_t)dW_t$$

- Reflecting barriers at $\underline{y}$ and $\bar{y}$
- Value function, distribution are now **functions of 2 variables**:

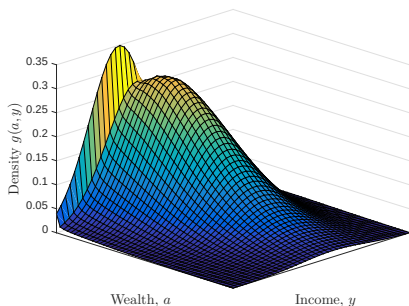
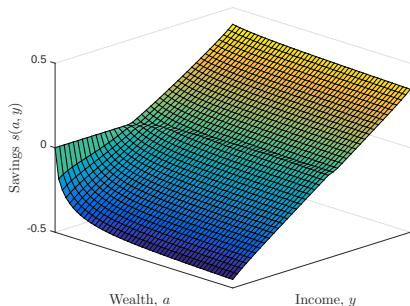
$$v(a, y) \quad \text{and} \quad g(a, y)$$

- $\Rightarrow$ HJB and KF equations are now **PDEs** in (a, y) -space

It doesn't matter whether you solve ODEs or PDEs
 $\Rightarrow$ everything generalizes

http://www.princeton.edu/~moll/HACTproject/huggett_diffusion_partialeq.m

Saving Policy Function and Stationary Distribution



- Analytic characterization of MPCs: $c(a, y) \sim \sqrt{2\nu(y)}\sqrt{a - \underline{a}}$ with

$$\nu(y) = (\rho - r)\text{IES}(\underline{c}(y))\underline{c}(y) + \left(\mu(y) - \frac{\sigma^2(y)}{2}\mathcal{P}(\underline{c}(y)) \right) \underline{c}'(y) + \frac{\sigma^2(y)}{2}\underline{c}''(y)$$

where $\mathcal{P}(c) := -u'''(c)/u''(c)$ = absolute prudence, and $\underline{c}(y) = c(\underline{a}, y)$

Other Applications – see Paper

- Non-convexities: indivisible housing, mortgages, poverty traps
- Fat-tailed wealth distribution
- Multiple assets with adjustment costs (Kaplan-Moll-Violante)
- Stopping time problems

Conclusion

- Very general apparatus: solving het. agent model = solving PDEs
- New theoretical results:
 1. analytics: consumption, saving, MPCs of the poor
 2. closed-form for wealth distribution with 2 income types
 3. unique stationary equilibrium if $IES \geq 1$
 4. characterization of “soft” borrowing constraints
- Computational algorithm:
 - simple, efficient, portable
 - codes: <http://www.princeton.edu/~moll/HACTproject.htm>
- Large number of potential applications – come talk to me!

Appendix

Derivation of Poisson KF Equation [▶ Back](#)

- Work with CDF (in wealth dimension)

$$G_j(a, t) := \Pr(\tilde{a}_t \leq a, \tilde{y}_t = y_j)$$

- Income switches from y_j to y_{-j} with probability $\Delta\lambda_j$
- Over period of length Δ , wealth evolves as $\tilde{a}_{t+\Delta} = \tilde{a}_t + \Delta s_j(\tilde{a}_t)$
- Similarly, answer to question “where did $\tilde{a}_{t+\Delta}$ come from?” is

$$\tilde{a}_t = \tilde{a}_{t+\Delta} - \Delta s_j(\tilde{a}_{t+\Delta})$$

- Momentarily ignoring income switches and assuming $s_j(a) < 0$

$$\Pr(\tilde{a}_{t+\Delta} \leq a) = \underbrace{\Pr(\tilde{a}_t \leq a)}_{\text{already below } a} + \underbrace{\Pr(a \leq \tilde{a}_t \leq a - \Delta s_j(a))}_{\text{cross threshold } a} = \Pr(\tilde{a}_t \leq a - \Delta s_j(a))$$

- Fraction of people with wealth below a evolves as

$$\begin{aligned} \Pr(\tilde{a}_{t+\Delta} \leq a, \tilde{y}_{t+\Delta} = y_j) &= (1 - \Delta\lambda_j) \Pr(\tilde{a}_t \leq a - \Delta s_j(a), \tilde{y}_t = y_j) \\ &\quad + \Delta\lambda_j \Pr(\tilde{a}_t \leq a - \Delta s_{-j}(a), \tilde{y}_t = y_{-j}) \end{aligned}$$

- Intuition: if have wealth $< a - \Delta s_j(a)$ at t , have wealth $< a$ at $t + \Delta$ ⁴⁷

Derivation of Poisson KF Equation

- Subtracting $G_j(a, t)$ from both sides and dividing by Δ

$$\frac{G_j(a, t + \Delta) - G_j(a, t)}{\Delta} = \frac{G_j(a - \Delta s_j(a), t) - G_j(a, t)}{\Delta} - \lambda_j G_j(a - \Delta s_j(a), t) + \lambda_{-j} G_{-j}(a - \Delta s_{-j}(a), t)$$

- Taking the limit as $\Delta \rightarrow 0$

$$\partial_t G_j(a, t) = -s_j(a) \partial_a G_j(a, t) - \lambda_j G_j(a, t) + \lambda_{-j} G_{-j}(a, t)$$

where we have used that

$$\begin{aligned} \lim_{\Delta \rightarrow 0} \frac{G_j(a - \Delta s_j(a), t) - G_j(a, t)}{\Delta} &= \lim_{x \rightarrow 0} \frac{G_j(a - x, t) - G_j(a, t)}{x} s_j(a) \\ &= -s_j(a) \partial_a G_j(a, t) \end{aligned}$$

- Intuition: if $s_j(a) < 0$, $\Pr(\tilde{a}_t \leq a, \tilde{y}_t = y_j)$ increases at rate $g_j(a, t)$
- Differentiate w.r.t. a and use $g_j(a, t) = \partial_a G_j(a, t) \Rightarrow$

$$\partial_t g_j(a, t) = -\partial_a [s_j(a, t) g_j(a, t)] - \lambda_j g_j(a, t) + \lambda_{-j} g_{-j}(a, t)$$

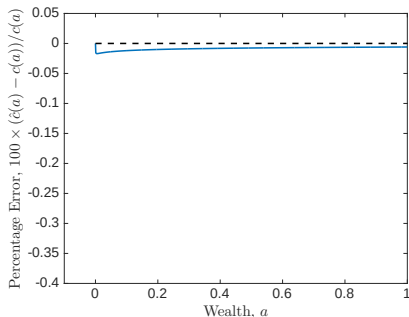
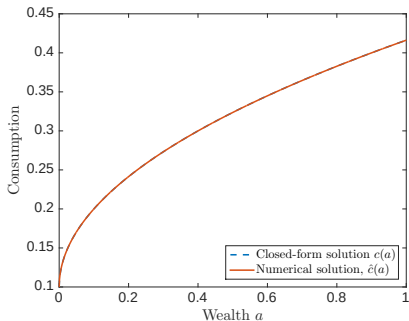
Accuracy of Finite Difference Method?

Two experiments:

1. special case: comparison with closed-form solution
2. general case: comparison with numerical solution computed using very fine grid

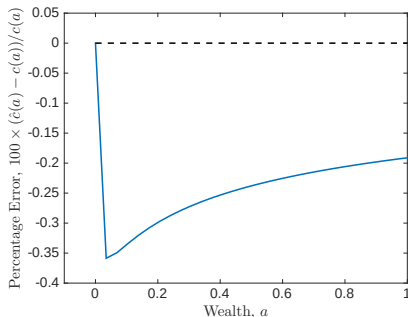
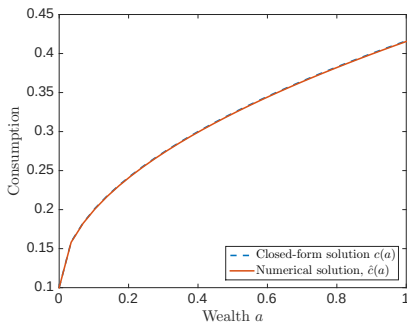
Accuracy of Finite Difference Method, Experiment 1

- See http://www.princeton.edu/~moll/HACTproject/HJB_accuracy1.m
- Recall: get closed-form solution if
 - exponential utility $u'(c) = c^{-\theta c}$
 - no income risk and $r = 0$ so that $\dot{a} = y - c$ (and $a \geq 0$)
 $\Rightarrow \quad s(a) = -\sqrt{2\nu a}, \quad c(a) = y + \sqrt{2\nu a}, \quad \nu := \frac{\rho}{\theta}$
- Accuracy with $I = 1000$ grid points ($\hat{c}(a)$ = numerical solution)



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- Accuracy with $I = 30$ grid points ($\hat{c}(a)$ = numerical solution)



Accuracy of Finite Difference Method, Experiment 2

- See http://www.princeton.edu/~moll/HACTproject/HJB_accuracy2.m
- Consider HJB equation with continuum of income types

$$\rho v(a, y) = \max_c u(c) + \partial_a v(a, y)(y + ra - c) + \mu(y) \partial_y v(a, y) + \frac{\sigma^2(y)}{2} \partial_{yy} v(a, y)$$

- Compute twice:
 1. with very fine grid: $I = 3000$ wealth grid points
 2. with coarse grid: $I = 300$ wealth grid points

then examine speed-accuracy tradeoff (accuracy = error in agg C)

	Speed (in secs)	Aggregate C
$I = 3000$	0.916	1.1541
$I = 300$	0.076	1.1606
row 2/row 1	0.0876	1.005629

- i.e. going from $I = 3000$ to $I = 300$ yields $> 10\times$ speed gain and 0.5% reduction in accuracy (but note: even $I = 3000$ very fast)
- Other comparisons? Feel free to play around with HJB_accuracy2.m